

LPIE Final Presentation Speaker Notes

Loan Performance Intelligence Engine (LPIE)

Intain FinTech AI Challenge 2026 - Final Round 3

Team: Narendra's Team | Sequence #9

Date & Slot: September 17, 2026 | 04:00 PM - 07:00 PM

Total Presentation Time: ~10 Minutes

Structure: 7 Minutes Slide Deck + 2.5 Minutes Live App Demo + 0.5 Minute Wrap-Up

QUICK PRESENTATION TIPS:

1. Speak with steady pacing; pause 1-2 seconds after stating key metrics (0.8168 AUC, 97.66 DQI).
2. Keep the live Streamlit tab pre-loaded in your browser for an instant tab switch during Slide 11.
3. Emphasize "Out-of-Time Validation" and "Competing Risks" - judges care deeply about financial rigor.

SLIDE 1: Title & Introduction

Target Timing: 30 seconds

What to Say:

"Good afternoon respected judges and the Intain team. I am Narendra, presenting the Loan Performance Intelligence Engine - LPIE - on behalf of Narendra's Team, Sequence Number 9.

LPIE is an enterprise mortgage risk surveillance platform that transforms raw monthly loan tapes into calibrated risk probabilities, competing-risk hazard curves, and audit-ready credit notes in under 40 seconds.

Let's examine the industry problem we are solving."

Transition: "Why does traditional secondary mortgage surveillance fail today?"

SLIDE 2: The Industry Problem vs LPIE Solution

Target Timing: 45 seconds

What to Say:

"Secondary mortgage surveillance today faces four critical flaws:

1. Delayed Surveillance: Manual rule reviews are too slow, missing 30 to 40 percent of early-stage delinquencies.
2. Statistical Bias: Traditional credit models treat default and prepayment as independent events. But a loan that prepays can never default. Prepayment censors default. Ignoring this creates severe positive survival bias.
3. Silent Servicer Errors: Data entry errors - like negative balances or impossible DPD jumps - slip undetected into securitization tapes.
4. LLM Hallucinations: Generic generative chatbots invent credit facts with zero auditability.

LPIE reverses this entirely: shifting from manual spreadsheets to calibrated ML on 34,000 panel records, competing-risks CIF survival curves, hybrid anomaly scanning, and 100% SHAP-grounded reviewer notes."

Transition: "Here is our modular end-to-end architecture."

SLIDE 3: 9-Phase End-to-End Pipeline Architecture

Target Timing: 60 seconds

What to Say:

"LPIE executes a modular, 9-phase automated pipeline.

We ingest 34,285 monthly panel records across 3,000 loans spanning 2022 to 2024 vintages. Phase 1 calculates our automated Data Quality Index. Phase 2 engineers 12 domain features while strictly quarantining 6 post-event administrative columns to eliminate target leakage.

We train calibrated XGBoost and LightGBM models evaluated on a strict out-of-time chronological partition - training on data up to October 2023, and validating on November 2023 through April 2024. No loan ID ever crosses the boundary.

This feeds directly into our Competing-Risks Survival engine, Hybrid Anomaly scanner, TreeSHAP explainability engine, and Grounded Reviewer Copilot.

The entire workflow is one-command reproducible: running ``python run_pipeline.py`` executes Phases 0 through 8 sequentially in just 40.3 seconds."

Transition: "Let's examine Phase 1: Data Intelligence."

SLIDE 4: Phase 1: Data Intelligence & DQI Profiling

Target Timing: 45 seconds

What to Say:

"For Task 1, our automated profiler scores the portfolio at 97.66 out of 100, achieving a certified Grade A data quality rating.

Completeness scored 97.4%. We employ median imputation coupled with explicit missingness indicator columns so we never lose statistical power.

Crucially, we evaluated out-of-time drift using the Population Stability Index (PSI). As shown in the Drift PSI chart on the right of this slide, every demographic and credit feature registered a PSI well below 0.04 - confirming zero distribution drift between our training and validation windows.

We also ran Pearson correlation matrices and Cramer's V tests to systematically audit multicollinearity and data integrity."

Transition: "With clean, validated data, let's examine our predictive models."

SLIDE 5: Phase 2: Multi-Horizon Calibrated Predictive Models

Target Timing: 60 seconds

What to Say:

"This is our core predictive engine - Task 2, worth 20 points.

Our headline result: 0.8168 ROC-AUC on 12-Month Default, representing a +12.6% lift over our baseline logistic regression model. On 12-Month Prepayment, we achieved 0.7119 ROC-AUC, a +23.8% lift.

Three methodological choices set our modeling apart:

First, strict out-of-time chronological validation: train window 2022-01 to 2023-10 versus validation window 2023-11 to 2024-04. No future information or loan ID overlaps.

Second, isotonic probability calibration: as shown in our reliability curves on the right, our predicted probabilities hug the diagonal 45-degree line, achieving a Brier Score of just 0.1063 - 43% lower error than baseline.

Third, an automated leakage audit verifying zero post-event administrative columns exist in our feature matrix."

KEY PHRASE TO EMPHASIZE: "Out-of-time chronological validation - zero loan contamination."

Transition: "Now, why standard classification is not enough - our Competing Risks Survival Engine."

SLIDE 6: Phase 3: Competing Risks Survival Engine

Target Timing: 60 seconds

What to Say:

"In mortgage finance, default and prepayment are mutually exclusive absorbing states. A loan that prepays can never default. Standard binary models treat prepaid loans as non-defaults, creating severe positive survival bias.

We solved this using Cumulative Incidence Functions (CIF), decomposing the 3,000-loan cohort into:

- 12.6 percent defaults (378 loans)
- 55.7 percent voluntary prepayments (1,672 loans)
- 31.7 percent active right-censored loans

Our statistical findings are definitive:

- The Log-Rank test proves Prime versus Subprime survival curves diverge with $p = 2.83 \times 10^{-13}$.
- In our Cox Proportional Hazards model, Prime credit tier confers a 26.9% reduction in monthly default hazard (HR = 0.731).
- Conversely, every 1% increase in interest rate accelerates default hazard by +57.8% (HR = 1.578).
- And our Weibull AFT model outperforms constant-hazard exponential models by 149.8 AIC points, proving hazard risk accelerates with loan seasoning."

KEY PHRASE TO EMPHASIZE: "Prepayment censors default - you must model them as competing risks."

Transition: "Next, how we catch anomalies before they corrupt portfolio valuations."

SLIDE 7: Phase 4: Hybrid Anomaly Detection

Target Timing: 45 seconds

What to Say:

"Phase 4 combines two complementary layers into a hybrid anomaly detector:

Layer 1 uses 10 deterministic accounting rules catching concrete violations: negative balances, current balance exceeding origination amount, impossible status transitions - like jumping from Current to 90 DPD in 30 days - and stale un-serviced records.

Layer 2 applies an unsupervised Isolation Forest for spatial density estimation, flagging multi-dimensional anomalies that pass individual rules but are statistically abnormal.

Our engine automatically compiled 25 reviewer-ready diagnostic case studies in reports/anomaly_report.md with explicit remediation actions for servicer reconciliation."

Transition: "What happens under macroeconomic stress? We simulate that in Phase 5."

SLIDE 8: Phase 5: Macroeconomic Stress Simulation

Target Timing: 45 seconds

What to Say:

"In Phase 5, we stress-test the entire portfolio across three macro regimes:

Under Baseline, the 12-month default rate is 14.83% and prepayment is 50.75%.

Under the Adverse Credit Scenario - simulating a +150 basis point rate shock combined with credit tightening - default rate surges by 129% to 33.91%, while prepayment drops 40% to 30.53%.

Under the High Prepayment Scenario - simulating a 100 bps rate cut - default drops to 9.81%, but 100% of the portfolio prepays as borrowers refinance.

This proves an essential secondary market insight: adverse credit regimes trigger asymmetric tail risk, where defaults surge while duration extends as prepayment dries up."

Transition: "How do we make all of these predictions transparent and legally compliant?"

SLIDE 9: Phase 6: TreeSHAP Explainability & Responsible AI

Target Timing: 45 seconds

What to Say:

"Every prediction in LPIE is fully explainable using TreeSHAP.

We provide three explainability dimensions:

1. Global Attribution: As shown in our beeswarm summary plot on this slide, `days_past_due`, `balance_to_orig_ratio`, and `interest_rate` emerge as the primary drivers of 12-month default across 1,000 validation samples.
2. Local Waterfall Audits: For any individual loan, we show exact dollar-for-dollar and percentage point contributions.
3. Error Diagnostics: We analyzed our False Positives and False Negatives to uncover systematic model edge cases.

This architecture is strictly ECOA and Regulation B Fair Lending compliant - no protected class attributes are ingested, and every credit decision is legally defensible with human-interpretable Shapley values."

Transition: "Now, our Reviewer Copilot."

SLIDE 10: Phase 7: Grounded Reviewer Copilot

Target Timing: 45 seconds

What to Say:

"Most AI reviewer tools use generative LLMs that frequently hallucinate credit facts. We took an anti-hallucination, deterministic approach.

Our Reviewer Copilot operates on a 3-step grounding protocol:

1. It ingests borrower attributes verified strictly against our Data Dictionary.
2. It computes local TreeSHAP attribution values for that exact loan.
3. It generates a structured reviewer note where every single qualitative claim MUST cite a numerical SHAP value.

If a feature doesn't have an exact SHAP attribution, the copilot will not state it. We verified this with 3 explicit rejection and correction test cases in `prompt_log.jsonl`, establishing an immutable regulatory audit trail."

KEY PHRASE TO EMPHASIZE: "Zero hallucination - every risk claim cites a numerical SHAP value."

Transition: "Let's look at deployment and switch to our live demo."

SLIDE 11: Phase 8: Production Deployment & Live Demo

Target Timing: 2.5 minutes

What to Say:

"LPIE is not a Jupyter notebook prototype - it is a production microservice suite.

It features a live Streamlit Cloud dashboard, a FastAPI REST server with interactive Swagger docs at /docs, full Docker containerization, and a 100% passing automated test suite (5/5 tests).

Let me switch to our live browser to show you the engine in action."

Live Demo Steps (on loan-performance-intelligence-engine.streamlit.app):

1. Mission Control (30s): Show live KPIs (34k records, 97.66 DQI) and donut distribution charts.
2. Real-Time Scoring (45s): Set Balance to \$285k, Rate to 7.5%, DPD to 60 -> click Calculate Risk -> show gauge meters & ESCALATE badge.
3. Stress Lab (30s): Drag Interest Rate Shift slider to +150 bps -> point out the live default rate surge to 33.91%.
4. Copilot (30s): Select loan -> click Generate Reviewer Note -> point out exact TreeSHAP citations in the text.

SLIDE 12: Conclusion & Rubric Victory Summary

Target Timing: 45 seconds

What to Say:

"To conclude, LPIE delivers a comprehensive, 100-point rubric-compliant solution across all 8 challenge dimensions:

- Phase 1: 97.66 DQI Score - Enterprise-grade data quality
- Phase 2: 0.8168 ROC-AUC - Out-of-time calibrated predictive excellence
- Phase 3: Competing Risks CIF - Statistically proven survival separation ($p = 2.83 \times 10^{-13}$)
- Phase 4: 25 Diagnosed Case Studies - Automated anomaly triage
- Phase 5: Interactive Stress Testing - Asymmetric tail-risk quantification
- Phase 6: 100% Explainable TreeSHAP - ECOA / Reg B Fair Lending compliance
- Phase 7: Zero-Hallucination Copilot - SHAP-grounded audit notes
- Phase 8: 40-Second Reproducible Pipeline - Live on Streamlit Cloud and FastAPI

LPIE transforms secondary mortgage surveillance from a slow, manual chore into an instant, auditable intelligence workflow.

Thank you, judges. Narendra's Team is ready for your questions!"